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Implementation & Creativity All tokenizers use word-level Byte Pair Encoding (BPE) with whitespace normalization and a `_` space marker. Key innovations:

- **Decoupled Encoding:** Replaced $O(W^2)$ merge-replay with an $O(W)$ greedy longest-match prefix trie over the trained vocab. This boosted encoding speed by +54% (Domain 1) and +64% (Domain 2) with negligible F1 dips (~ 0.001 – 0.009).
- **Length-Preserving [UNK]:** Maps unseen characters to a single symbol, preventing downstream NER character-span misalignment.
- **Bounded Bigrams:** Forces the top adjacent word-pairs (e.g., `_in_the`) into tokens, strictly capping at two words (≤ 1 internal space).

Parameter Selection & Validation Per course staff approval, we utilized per-tokenizer vocabularies to optimize the efficiency-F1 trade-off.

- **Model 1 (Domain 1 - Twitter):** Highly noisy (67.8% OOV entities). Smaller vocabularies force rare entities into shared subwords, vastly improving generalization. We locked **Vocab = 2000**.
- **Model 2 (Domain 2 - News):** Cleaner text, but heavily fragmented by rare unicode characters. F1 is robust, allowing **Vocab = 5000** for a 21% efficiency gain. (*A sweep of `num_bigrams` yielded negligible gains and increased fragmentation, remaining locked at 5*).

Final Dev Set Validation (Longest-Match Encoder):

- **Model 1 (D1, Vocab 2000):** F1 = 0.4597 | Efficiency = 0.337 tok/char | Speed = 560k tok/s
- **Model 2 (D2, Vocab 5000):** F1 = 0.9601 | Efficiency = 0.283 tok/char | Speed = 374k tok/s

Bigram Frequencies (Train Set)

- **Domain 1:** *Top:* `_in_the`, `_for_the`, `_going_to`, `_to_be`, `_on_the`. *Bottom:* `_hockey_3-0.`, `_ta_so`, `_Queen_will`, `_voice_as`, `@officialSusanB_ta`.
- **Domain 2:** *Top:* `_of_the`, `_in_the`, `_to_the`, `_on_the`, `_to_be`. *Bottom:* `_a_Wonder`, `_10-year-old_self`, `_Kids:_Reflections`, `_Reflections_During`, `Giving_Culture..`

Hidden Domain Strategy (Tokenizer 3) Tokenizer 3 trains on both `domain_1` and `domain_2`.

Downsampling D1 to balance the data degraded Dev F1, proving that maximizing raw data outpaces de-biasing. We assigned T3 **Vocab = 5000** for efficiency. We expect strong hidden-domain F1: T3's D1-proxy F1 (0.4879) actually surpassed T1 (0.4597). The length-preserving [UNK] and shared subwords ensure all unseen entities retain alignment-safe subtokens.

